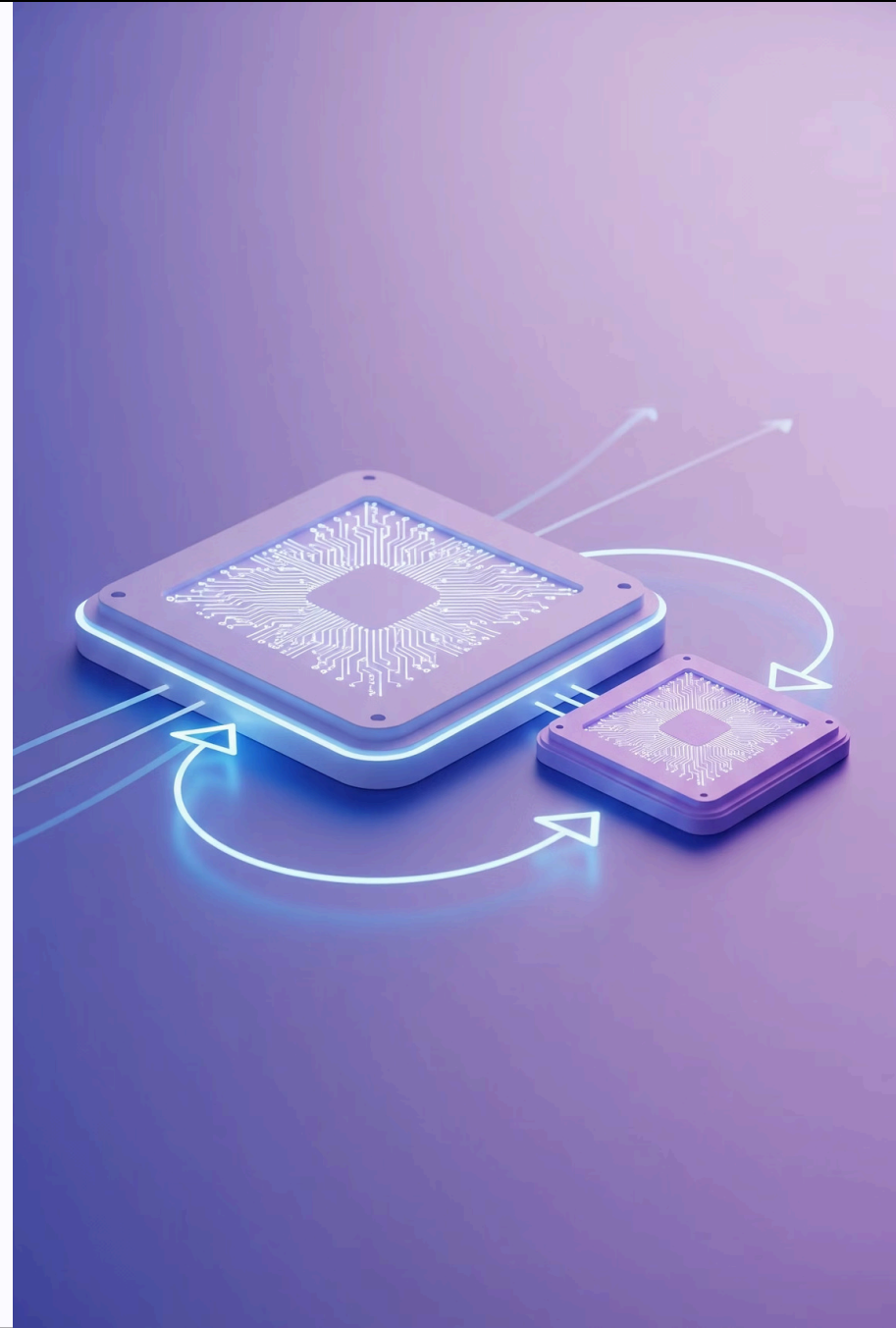


Distillation Basics: Teaching Smaller Models to Mimic Larger Ones

A practical approach to creating efficient AI models while maintaining performance

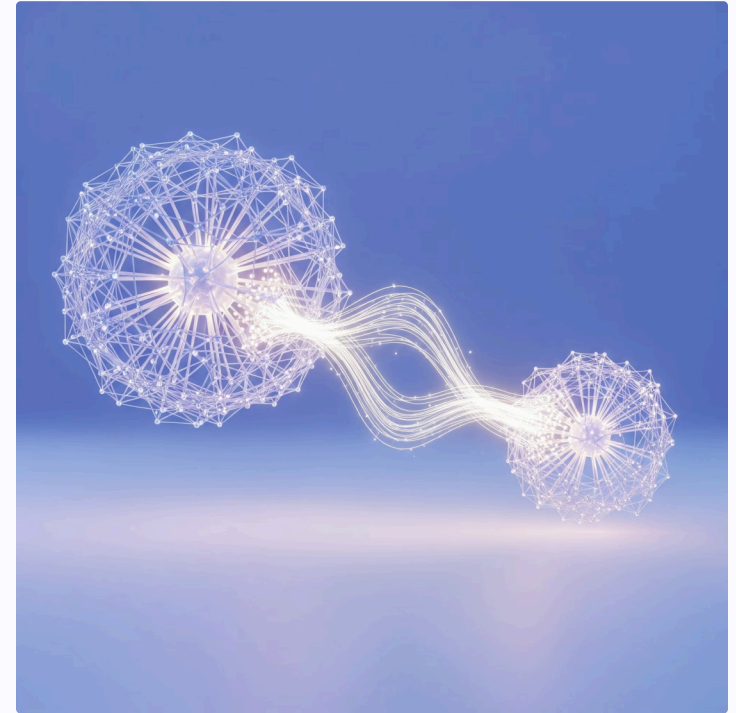


What Is Knowledge Distillation?

Knowledge distillation is the process of transferring knowledge from a **large teacher model** to a **smaller student model**.

The student learns not just hard labels, but also **soft predictions** (logits/probabilities) from the teacher.

The ultimate goal: Create a **smaller, faster, cheaper** model that maintains near-teacher accuracy.



Why Distill?

Edge & Mobile Deployment

Large models are often too big for **edge/mobile/real-time** applications where resources are limited.

Cost Efficiency

Serving costs scale with model size (GPU/TPU hours).
Smaller models = lower operational costs.

Performance Preservation

Distilled students are **compact but accurate**, maintaining much of the teacher's capabilities.

Democratizing AI

Key to making frontier models accessible to organizations with limited computing resources.

Core Concepts



Teacher Model

Pre-trained, large, highly accurate model (e.g., BERT, GPT)



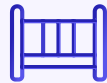
Student Model

Smaller, lightweight architecture (e.g., DistilBERT)



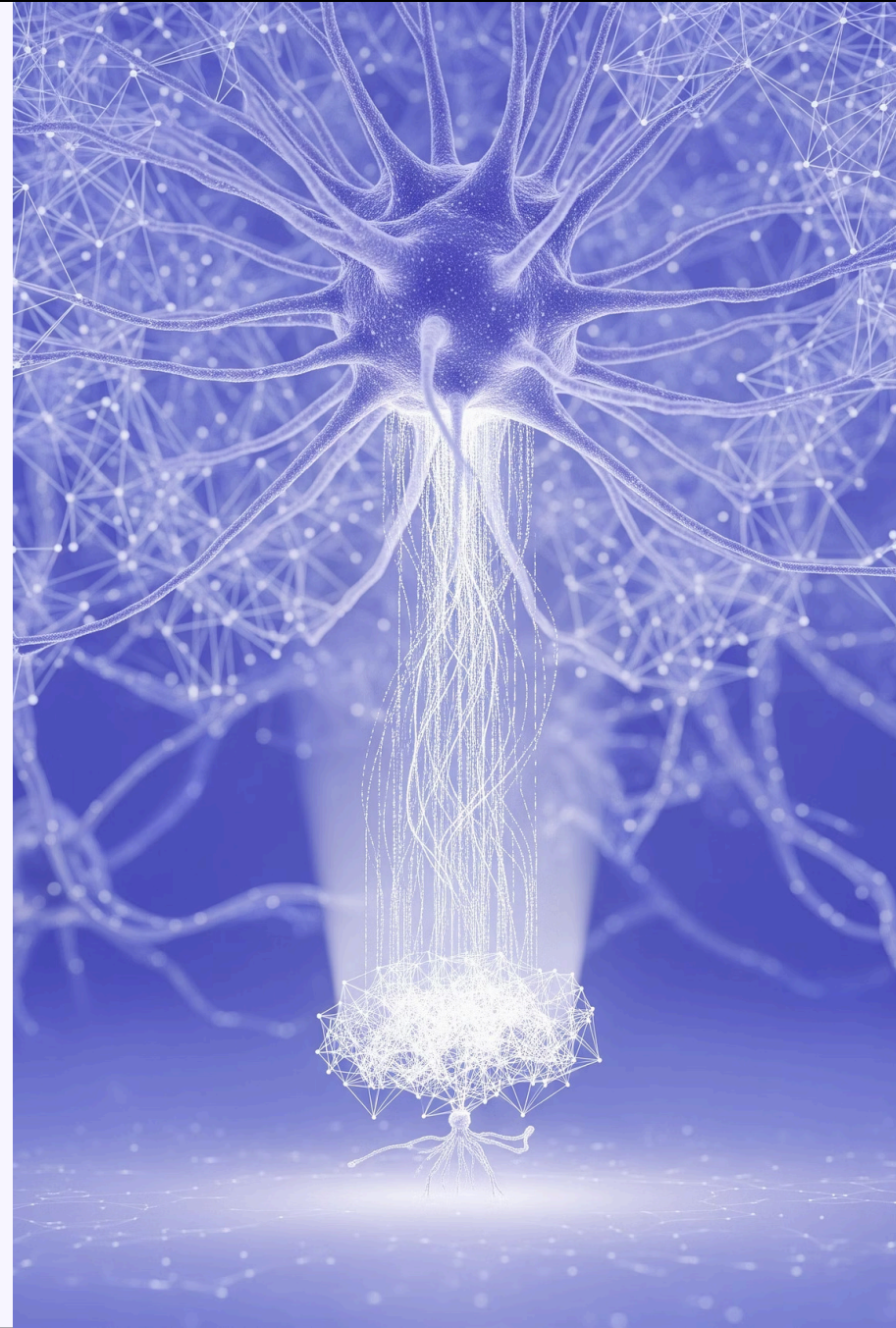
Soft Targets

Teacher output probabilities that guide student learning



Distillation Loss

Mix of hard-label loss + soft-label loss from teacher



Types of Distillation



Logit Distillation

Student learns to mimic teacher's softmax outputs - the most common approach



Feature Distillation

Aligns intermediate representations between teacher and student models



Attention Distillation

Student learns to reproduce attention maps (especially common in Transformers)



Self-Distillation

Model iteratively teaches improved versions of itself

Distillation Loss Function

Student learns from both ground truth labels and teacher's soft predictions:

$$L = (1 - \alpha) \cdot CE(y_{true}, y_{student}) + \alpha \cdot T^2 \cdot KL(p_{teacher}(T), p_{student}(T))$$

Key Components:

- T = temperature parameter to soften probability distributions
- α = weight that balances true labels vs. teacher guidance
- CE = cross-entropy loss with ground truth
- KL = Kullback-Leibler divergence between distributions



Example: Hugging Face DistilBERT

```
from transformers import DistilBertForSequenceClassification

# Pretrained DistilBERT already distilled from BERT
student = DistilBertForSequenceClassification.from_pretrained(
    "distilbert-base-uncased"
)

student.train() # fine-tune on your dataset
```

DistilBERT is a popular example of knowledge distillation applied to BERT, resulting in a model with 40% fewer parameters while retaining 97% of BERT's performance.



Benefits of Knowledge Distillation

50%+

Size Reduction

Significant parameter count reduction compared to teacher models

2-5×

Inference Speedup

Faster prediction time for real-time applications

↓ 60%

Memory Usage

Lower memory footprint enables deployment on resource-constrained devices

↓ 70%

Energy Consumption

Reduced energy requirements for more sustainable AI

Distilled models maintain most of the teacher's accuracy while dramatically improving efficiency metrics.

Limitations

- Performance Gap

Student never fully matches teacher accuracy - expect some degradation

- Training Overhead

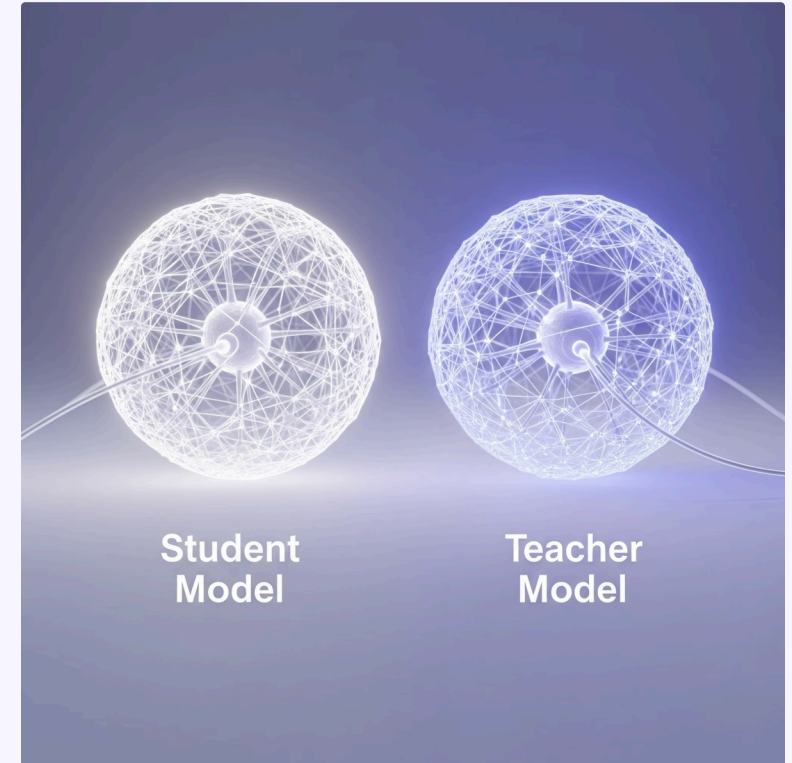
Distillation process adds initial training cost and complexity

- Hyperparameter Sensitivity

Results highly dependent on choice of teacher and temperature settings

- Bias Inheritance

Students may **inherit biases** present in teacher models



Real-World Examples



DistilBERT

40% smaller, 60% faster than BERT, while retaining 97% of its performance on language understanding tasks



TinyBERT & MobileBERT

Specially optimized for edge deployment with extreme compression techniques



LLM Distillation

Mini-GPT, Phi-2, and Alpaca models derived from larger teachers like GPT-3 and LLaMA

Best Practices

1

Teacher Selection

Choose a **well-trained teacher** - garbage in → garbage out

2

Temperature Tuning

Carefully tune **temperature (T)** parameter for optimal soft-label learning

3

Task-Specific Data

Fine-tune student on **task-specific dataset** after distillation

4

Complementary Techniques

Combine with **quantization/pruning** for maximum efficiency gains



Careful hyperparameter tuning and architectural decisions can significantly improve distillation results.

Key Takeaways

Knowledge Transfer

Distillation teaches a small model to mimic a larger one by learning from its probability distributions

Efficiency vs. Performance

Dramatically cuts size/latency while preserving most of the accuracy of larger models

Practical Deployment

Critical for edge, mobile, and cost-efficient inference in production environments

Future of AI

Core method in **efficient AI at scale** - will become increasingly important as model sizes grow

- ❏ Next Steps: Explore distillation techniques for your specific models and deployment constraints. Consider the trade-offs between performance and efficiency for your use case.